CARTHAGENTS CHALLENGE · ZERO-HALLUCINATION AUDIT AGENT

A VLM reads the layout. Code reads every figure. Accounting laws decide.

NEURAL · gpt-5.6-sol

Understands the page

Labels columns, rows, unit and period. Answers with ids and labels only — its schema has no place for a number.

CODE · text layer + docTR OCR

Copies the figures

Reads the printed characters in each row × column box, parses them to cents, keeps a box and a confidence per value.

SYMBOLIC · accounting laws

Decides what to trust

Accepts the VLM's reading only where the roll-forwards hold. Unsure? Blank and flag: +1 · 0 · −3.

PDF in (text, scan, outline) → filled Arbeitsversion + one JSON record per value with page, box and confidence

From printed characters to the Arbeitsversion

0 · TOKENS
Text layer, or docTR OCR after deskew and grid removal. Box + confidence per word.

1a · GEOMETRY
Columns from the numbers' right edges, one row band per printed line.

1b · VLM LABELS
Page image + column and row ids in; field, row kind, unit, period out.

2 · EXTRACTION
String in each box → cents. Ink check for empty OCR cells.

3 · VERIFICATION
Laws pick the mapping, totals by arithmetic, decide per value.

The laws that check everything

```
AHK end = start + additions - disposals ± transfers
kum. AfA end = start + AfA - AfA on disposals
- write-ups
book value = AHK - kum. AfA (now and prior year)
subtotal = sum of the rows it closes
```

Only the laws a document's printed columns allow are derived and checked.

Situation	Output
Value verified by a law or two readers	written + pointer
Client's subtotal does not add up	as printed + flag
OCR confidence < 0.75	blank + flag
Scanned row breaks a law	blank + flag
No law, VLM and lexicon disagree	blank + flag

Every step writes its own JSON file, so any value can be traced from its printed characters to its cell in the workbook.

The symbolic part: accounting laws as the judge

1 · DERIVE THE LAWS

Which checks can this document support?

Four identities hold in every schedule. Gaussian elimination keeps only the ones the printed columns allow, e.g. a net print still yields: book value = prior + additions − disposals ± transfers + write-ups − AfA.

2 · TEST THE VLM

Is the column mapping true?

The VLM's mapping and the alternatives from the header words are scored on every row. The one under which the laws hold wins; a swapped or shifted column breaks them and loses.

3 · TOTALS BY ARITHMETIC

Which rows are sums?

A total equals the rows above it not yet summed, labelled or not, above or below its rows. The grand total closes everything; what follows it is not the schedule.

4 · DECIDE

A value is written only if code copied it **and** a law (or two independent readers) confirms what it means. A client's broken sum is written as printed and flagged. Anything unverified is blank with a reason: **+1 · 0 · −3**.

The laws are accounting facts, not layout rules — so the same check works on layouts nobody has seen.

Results so far and what comes next

+9,080

Arbeitsversion score on the 54 sample PDFs (was −685)

190 → 0

wrong values on three unseen generated layouts, header words vs VLM + laws

0.998

mean docTR confidence on amounts in a skewed, fully ruled scan

Arbeitsversion vs ground truth	correct	blank	incorrect	score
gpt-4.1 + tesseract, first writer	10,475	14,220	3,720	−685
+ template conventions	12,036	12,625	3,539	+1,419
gpt-5.6-sol + docTR	13,244	11,377	1,388	+9,080

Next

- Reader for records printed over several lines (7,199 blanks)
- Tune the OCR threshold on confidence vs outcome
- Score on held-out generated layouts, PDF only

1,254 of the 1,388 “incorrect” cells are numbers in rows the comparison could not pair (placement); 94 held a different number.